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Comptia N+ Assignment

[N+ Network Fundamental and building networks]

IP addressing & Subnetting

Task:

❖ **1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.**

➢ **Introduction**

- An IP (Internet Protocol) address is a logical address used to identify a device on a network and enable communication between devices. In the traditional IPv4 addressing system, IP addresses are divided into five classes: A, B, C, D, and E. Each class has a different address range and purpose.

➢ **IP Address Classes and Their Uses**

IP Class	First Octet Range	Default Subnet Mask	Purpose	Real-World Example

Class A	1–126	255.0.0.0 (/8)	Large networks	A large organisation using a private network such as 10.0.0.0/8.
Class B	128–191	255.255.0.0 (/16)	Medium-sized networks	A college network using a private address such as 172.16.0.0/16.
Class C	192–223	255.255.255.0 (/24)	Small networks	A home LAN using addresses such as 192.168.1.10.
Class D	224–239	Not applicable	Multicast communication	A router or streaming service using multicast to distribute video to multiple receivers.
Class E	240–255	Not applicable	Experimental and reserved purposes	Research or testing of experimental IP networking systems.

- **Explanation**

1. Class A: Designed for very large networks with a large number of possible host addresses.

2. Class B: Traditionally used for medium-sized networks, such as networks in colleges and large organisations.

3. Class C: Traditionally suitable for smaller networks, such as home and small-office LANs.

4. Class D: Used for multicast communication, where data is sent to a group of interested devices rather than to just one device.

5. Class E: Reserved for experimental or future-use purposes and is not generally used for ordinary host addressing.

❖ **2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Show your working steps for each calculation.**

➢ **Calculation of Network Address, Broadcast Address, and Valid Host Range**
➢ **Given:**
- IP Address: 192.168.10.45
- Subnet Mask: 255.255.255.0
- CIDR Notation: /24

➢ **1. Calculation of Network Address**
Formula:
Network Address = IP Address AND Subnet Mask
Working Steps:
➢ Convert the IP address and subnet mask into binary:
➢ IPAddress:
192.168.10.45
11000000.10101000.00001010.00101101
➢ SubnetMask:
255.255.255.0
11111111.11111111.11111111.00000000
➢ Perform the bitwise AND operation:
IP Address: 11000000.10101000.00001010.00101101

Subnet Mask: 11111111.11111111.11111111.00000000

Network: 11000000.10101000.00001010.00000000

- The result in decimal notation is **192.168.10.0**.
- **Network Address = 192.168.10.0**

- **2. Calculation of Broadcast Address**
- The broadcast address is obtained by setting all host bits to 1.
- Since the subnet mask is /24, the first 24 bits represent the network and the last 8 bits represent the host.
- **Working Steps:**
  Network bits: 11000000.10101000.00001010
  Host bits: 11111111
- Combine the network bits with all host bits set to 1:
  Broadcast: 11000000.10101000.00001010.11111111
- Convert the binary address into decimal notation:
  - 11000000 = 192
  - 10101000 = 168
  - 00001010 = 10
  - 11111111 = 255
- **Broadcast Address = 192.168.10.255**

- **3. Calculation of Valid Host Address Range**
- The valid host range includes all addresses between the network address and the broadcast address, excluding both.
- **Working Steps:**

- First valid host address = Network Address + 1
- Last valid host address = Broadcast Address − 1

➢ Therefore,

First valid host = 192.168.10.0 + 1
= **192.168.10.1**
Last valid host = 192.168.10.255 − 1
= **192.168.10.254**

➢ **Valid Host Address Range = 192.168.10.1 to 192.168.10.254**

➢ **4. Calculation of Total Valid Hosts**

The formula for the number of valid host addresses is:
[
\text{Valid Hosts} = 2^h - 2
]
Where (h) is the number of host bits.
Here, the number of host bits is:
[
h = 32 - 24 = 8
]
Therefore,
[
\text{Valid Hosts} = 2^8 - 2 = 256 - 2 = 254
]

➢ **Total Valid Host Addresses = 254**
➢ **Final Answer**

Particulars	Result
IP Address	192.168.10.45
Subnet Mask	255.255.255.0 (/24)
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Valid Host	192.168.10.1
Last Valid Host	192.168.10.254
Total Valid Hosts	254

❖ **3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.**

➢ Introduction

- Subnetting is the process of dividing a large network into smaller logical networks called subnets. It improves network management, security, and efficient utilization of IP addresses.
- A food delivery startup like Zomato requires separate subnets for Delivery Partners, Restaurants, Customers, and Admins to organize its network and help control access between different groups.

➢ Given Information

- **Given Network:** 10.0.0.0/24
- **Number of Required Subnets:** 4
- **Original Subnet Mask:** 255.255.255.0
- **Original Network Size:** 256 IP addresses

➢ Subnetting Calculation

➢ To create 4 subnets, we need to borrow 2 bits from the host portion.

➢ Formula:

$$
\text{Number of Subnets} = 2^n
$$

Where (n) is the number of borrowed bits.

$$
2^2 = 4 \text{ subnets}
$$

The new prefix length is:

[
/24 + 2 = /26
]
The new subnet mask is **255.255.255.192**.
Each /26 subnet contains:
[
2^{32-26} = 64 \text{ IP addresses}
]
Usable host addresses per subnet:
[
64 - 2 = 62 \text{ hosts}
]

- **Subnetting Plan**

Group	Network Address	Subnet Mask	Valid Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

- **Explanation of Each Subnet**

- **1. Delivery Partners Subnet:**
  - This subnet is assigned to delivery partners' devices and related internal systems. It provides 62 usable host addresses.
- **2. Restaurants Subnet:**
  - This subnet is used for restaurant-related devices and systems, helping keep restaurant network traffic separate from other groups.
- **3. Customers Subnet:**
  - This subnet is allocated to customer-facing devices or systems that require network access. In a real-world service, customers usually connect through the internet rather than sharing one private subnet directly.
- **4. Admins Subnet:**
  - This subnet is reserved for administrators and authorized staff. Access restrictions can be applied to protect sensitive company data and management systems.

- **Advantages of This Subnetting Plan**
  - Better Security: Separate subnets help isolate different groups.
  - Easy Management: Each group has its own network range.
  - Efficient IP Allocation: The /24 network is divided into four equal subnets.
  - Improved Organization: Network traffic can be managed through routing and access-control rules.

❖ **4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Show your calculation steps and final answers.**

➢ **Introduction**

- Subnetting is the process of dividing a large network into smaller networks called subnets. It helps improve network management, security, and efficient utilization of IP addresses.
- In a college Wi-Fi network, subnetting can be used to assign separate networks to different buildings, such as the library, laboratories, classrooms, and administrative offices.

➢ **Given Information**

- **Original Network Address:** 172.16.0.0/16
- **Original Subnet Mask:** 255.255.0.0
- **Minimum Required Subnets:** 20

➢ **Calculation of New Subnet Mask**

➢ To create at least 20 subnets, we need to borrow bits from the host portion.

➢ **Step 1: Calculate the number of borrowed bits.**

Formula:
[
\text{Number of Subnets} = 2^n
]
Where (n) is the number of borrowed bits.

We need at least 20 subnets:
[
2^4 = 16 \quad \text{(not enough)}
]
[
2^5 = 32 \quad \text{(sufficient)}
]
Therefore, **5 bits must be borrowed**.

- **Step 2: Calculate the new prefix length.**

Original prefix length = /16
Borrowed bits = 5
[
\text{New Prefix} = 16 + 5 = /21
]
Therefore, the new subnet prefix is **/21**.

- **Step 3: Calculate the new subnet mask.**
- The binary subnet mask for /21 is:
  11111111.11111111.11111000.00000000
- Converting it into decimal notation:
  255.255.248.0
- **New Subnet Mask = 255.255.248.0 (/21)**

- **Calculation of Hosts Available per Subnet**
- **Step 1: Calculate the remaining host bits.**

IPv4 addresses contain 32 bits.

$$
\text{Host Bits} = 32 - 21 = 11
$$

- **Step 2: Calculate the total IP addresses per subnet.**

Formula:

$$
\text{Total IP Addresses} = 2^h
$$

Where (h) is the number of host bits.

$$
2^{11} = 2048
$$

Therefore, each subnet contains **2048 total IP addresses**.

- **Step 3: Calculate the usable host addresses.**

Formula:

$$
\text{Usable Hosts} = 2^h - 2
$$

$$
= 2^{11} - 2
$$

$$
= 2048 - 2 = 2046
$$

Two addresses are excluded: the network address and the broadcast address.
Therefore, **usable host addresses per subnet = 2046**.

- **Final Answers**

Particulars	**Result**
Original Network	172.16.0.0/16
Original Subnet Mask	255.255.0.0
Required Subnets	At least 20
Bits Borrowed	5
Total Subnets Created	32
New Prefix Length	/21
New Subnet Mask	255.255.248.0
Host Bits Remaining	11
Total IP Addresses per Subnet	2048
Usable Hosts per Subnet	2046

❖ **5. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.**

➢ **Invalid IP Address and Its Correction**

➢ **Explanation of the Error**

- The IP address **192.168.1.300** is invalid because an IPv4 address consists of four octets separated by dots, and each octet must have a value between **0 and 255**.
- In the given IP address, the last octet is 300, which exceeds the maximum permitted value of 255. Therefore, the device cannot use this IP address for normal IPv4 network communication.

➢ **Example of a Valid IP Address**

➢ For a typical network using the subnet mask **255.255.255.0 (/24)**, a valid host IP address could be:
192.168.1.100

➢ In this subnet:

- Network Address: 192.168.1.0
- Valid Host Range: 192.168.1.1 to 192.168.1.254
- Broadcast Address: 192.168.1.255

➢ The IP address 192.168.1.100 is valid, provided it is not already assigned to another device.